

PROJECT BRIEF

Edge-Native Multimodal Meeting & Task Copilot

Private by design. Useful in real time. Built and optimized for Snapdragon®-powered HP PCs (HP OmniBook X) — where raw audio, screen and context never have to leave the device to be useful.

100% LOCAL AI

NPU-FIRST

ARM64

ZERO CLOUD DATA PATH

CHALLENGE

Snapdragon® AI Lab Build & Present

- 01 Problem & opportunity
- 02 Solution overview & pipeline
- 03 Technical architecture
- 04 Snapdragon & Qualcomm AI Hub alignment
- 05 Signature feature — Privacy Guard
- 06 Benchmark plan & measured evidence
- 07 Deployment & accessibility
- 08 Demo script
- 09 Execution plan & close

SUBMITTED BY

Siddharth Goyal • Individual Participant

TARGET DEVICE

HP OmniBook X (Snapdragon X Elite)

01 PROBLEM & OPPORTUNITY

The meeting copilot has a trust problem

Today, useful AI often requires sending the most sensitive part of work — voice, screen and context — somewhere else. Every cloud round trip is a privacy decision made on the user's behalf.



Cloud round trips

Live audio / screen → network → inference → response.
Performance varies with connectivity, and every hop is a new place data can be intercepted or logged.



Sensitive context

Credentials, customer records and internal documents can appear in a screen share unexpectedly — with no local safeguard to catch them before they're sent.



Recurring cost

Always-on transcription and LLM calls turn every meeting minute into metered cloud usage, scaling cost with usage instead of hardware already paid for.



Battery contention

Using CPU/GPU for continuous background AI competes with the actual meeting workload — video, screen share, and everything else running at once.

DESIGN QUESTION

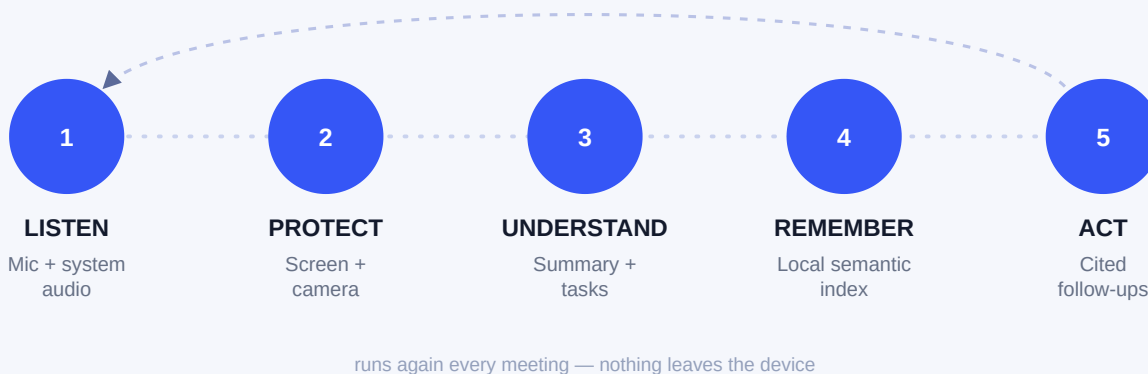
Can the PC itself become the privacy boundary — and still feel instant?

AegisLens is a bet that the answer is yes: the same NPU that Snapdragon X Elite ships for on-device AI is enough to run a full meeting-copilot pipeline — speech, vision, reasoning and memory — without a single byte leaving the machine.

02 SOLUTION OVERVIEW

One local loop instead of four cloud dependencies

AegisLens replaces the usual capture → cloud → response round trip with a single closed loop that runs entirely on the Snapdragon NPU, from the moment audio is captured to the moment a task is recalled weeks later.

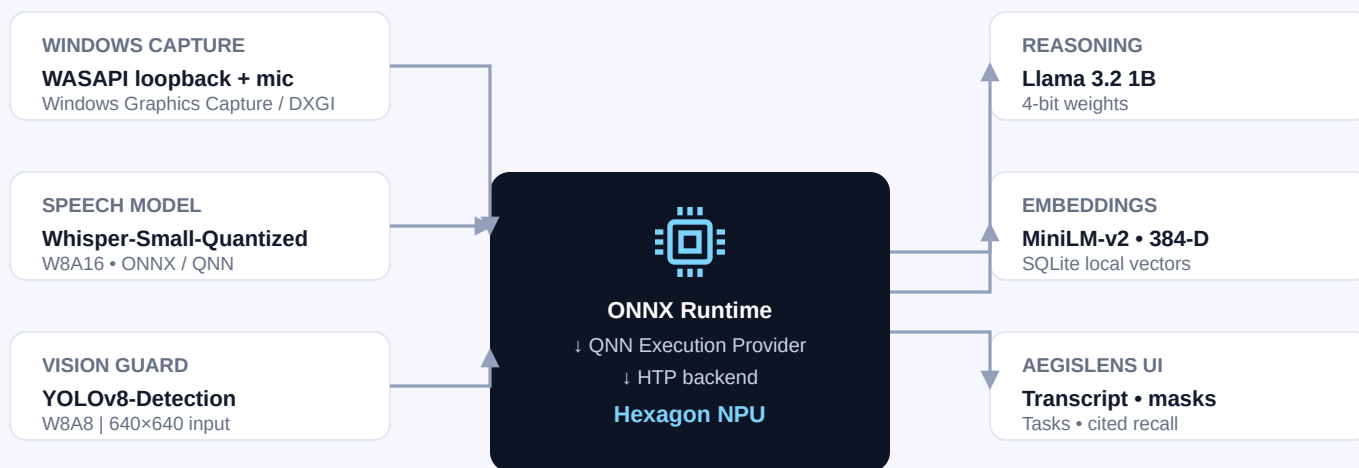


WHY THIS WINS

The differentiator is not "another meeting summary." It is a privacy-first pipeline where raw multimodal data can stay on-device from capture to recall — a claim that is architecturally true, not a policy promise.

Route each workload to the right engine

Capture stays close to the OS; every model runs through ONNX Runtime's QNN Execution Provider straight to the Hexagon NPU, so the CPU/GPU stay free for the meeting itself.



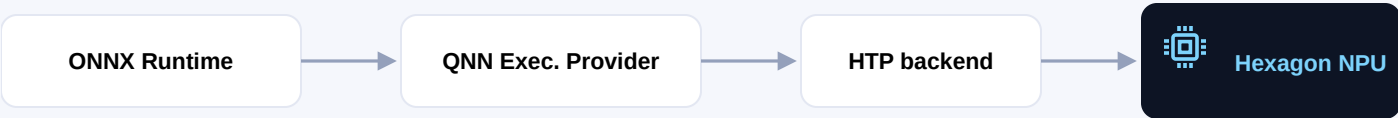
Frame path stays GPU-resident where possible; QNN context is cached to avoid repeated graph setup.

Qualcomm AI Hub is part of the build loop — not a logo

Every model on the critical path is profiled and exported through Qualcomm AI Hub, targeting the HP OmniBook X's Snapdragon X Elite NPU.

WORKLOAD	MODEL	AI HUB STEP	TARGET DEVICE
Speech-to-text	Whisper-Small-Quantized	W8A16 • profile/export	HP OmniBook X — NPU
Action extraction	Llama v3.2 1B Instruct	W4 + partial W8	HP OmniBook X
Privacy vision	YOLOv8-Detection	W8A8 • 3.25 MB	HP OmniBook X — NPU
Semantic recall	MiniLM-v2 384-D embeddings	—	HP OmniBook X

NPU offload path

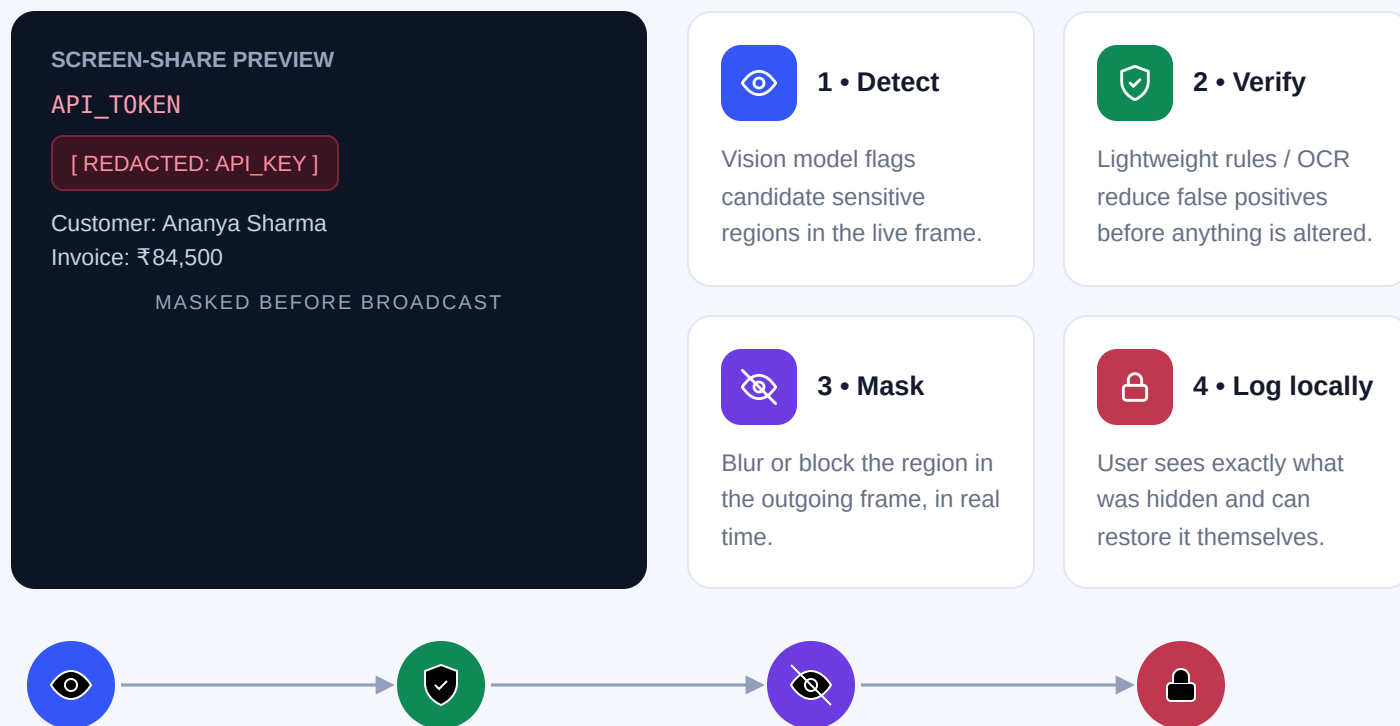


Runtime proof: AI Hub profile **jgd3j01lp** — Whisper-Small-Quantized encoder • Snapdragon X Elite CRD (Qualcomm compliance reference device) • Windows 11 • QNN EP • NPU. Final on-device validation targets the HP OmniBook X directly during the benchmark pass (Sep 27–28).

05 SIGNATURE FEATURE

Privacy Guard acts before pixels leave the device

A local vision model watches the outgoing screen-share frame, not the recording after the fact — so sensitive data never reaches a viewer, let alone a cloud service.

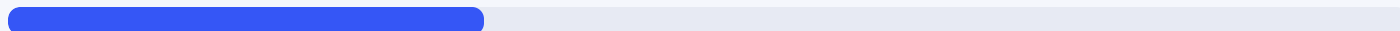


Fail-safe idea: when confidence is uncertain, warn the presenter instead of silently altering content.

Measure what judges can reproduce

Every number below is either a Qualcomm AI Hub profile output or an on-device measurement — each is labeled with exactly where it came from.

ASR encoder latency 305 ms



Qualcomm AI Hub • profile jgd3j01lp

App warm start 801 ms



On-device measurement

App cold start 890 ms



On-device measurement



127 MB

Peak ASR memory



Hexagon NPU

Primary accelerator



None

Cloud data path



Verified

Reproducible on
request

AI Hub metrics (encoder latency, memory, accelerator) come directly from Qualcomm AI Hub profile **jgd3j01lp** — Whisper-Small-Quantized encoder, Snapdragon X Elite CRD, Windows 11, QNN EP. App-level start-up times are measured on-device, separately from the AI Hub graph profile, and will be re-verified on the HP OmniBook X during the Sep 27–28 benchmark pass.

07 DEPLOYMENT & ACCESSIBILITY

Ship like a Windows app, not a research notebook

A judge should be able to install, understand and verify the core claim without reading the source code first.



Native ARM64

MSIX • Windows App SDK / WinUI 3 • native ARM64 • versioned model assets.



Graceful degradation

NPU preferred. If a model cannot offload, the fallback is surfaced clearly instead of hidden.



Accessible controls

Keyboard-first actions • captions • scalable text • high-contrast-safe UI.



Local storage

SQLite + local vector index • per-meeting retention • one-click purge.



Permission model

WASAPI + screen capture are explicit; no network permission in the production manifest.



Reproducible setup

README includes device, runtime, model version, QNN config and benchmark steps.



HP AI ECOSYSTEM ALIGNMENT

Built for the HP OmniBook X's Snapdragon X Elite platform — a 45 TOPS NPU and HP-published battery life of up to ~26–28 hours. Continuous background capture is designed to run entirely on the NPU to stay within that headroom; actual thermal and battery impact will be measured on-device during the Sep 27–28 benchmark pass, not asserted in advance.

08 DEMO SCRIPT

A demo judges can understand in 90 seconds

Every beat proves a specific claim on screen — nothing is asserted that isn't shown.

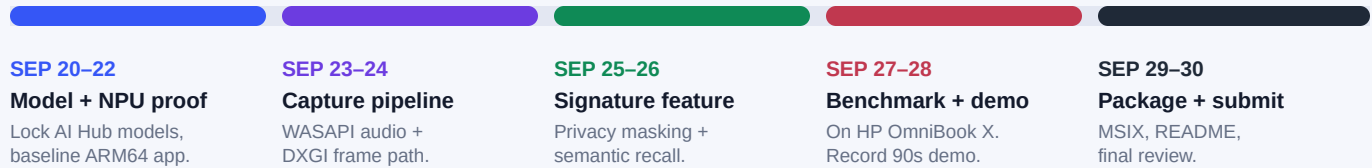
- 1 00:00 Start meeting** AegisLens shows the LOCAL / NPU badge — proves the pipeline is live and on-device from second one.
- 2 00:15 Say a commitment** "I'll send the revised proposal Friday." → a task appears automatically, cited to the moment it was said.
- 3 00:35 Share a secret** Open a mock API key on screen → Privacy Guard masks it before the frame is broadcast.
- 4 00:55 Ask from history** "What did I promise?" → the answer cites the exact transcript timestamp, pulled from the local index.
- 5 01:15 Open diagnostics** Show the QNN execution provider, model names and live resource counters — the receipts behind every claim above.



PROTOTYPE REPOSITORY

github.com/Siddharth713-cpu/aegislens

Build plan to September 30



WHAT MAKES AEGISLENS CREDIBLE

The strongest claims are designed to be demonstrated, measured and reproduced — not asserted on a slide.



100% on-device
inference



Privacy Guard, live



Cited, verifiable recall



Built for HP OmniBook
X